INCH-POUND

MIL-C-47224B (MI)
4 JANUARY 1993
SUPERSEDING
MIL-C-47224A (MI)
20 February 1981

MILITARY SPECIFICATION

COMPOUND, MOLDING, TRANSFER, EPOXY RESIN, SINGLE-COMPONENT

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of single-component, epoxy resin transfer molding compound for encapsulating, potting, and embedding electrical devices requiring good thermal conductivity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330
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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-96	-	Cans, Metal, 28 Gauge and Lighter
PPP-D-705	-	Drum, Shipping and Storage: Steal, 16 and 30 Gal. Capacity
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liter)

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-I-16923	-	Insulating Compound, Electrical, Embedding

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of

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documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 149	-	Standard Test Methods for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies
ASTM D 150	-	Standard Test Methods for A-C Loss Characteristics and Permitivity (Dielectric Constant) of Solid Electrical Insulating Materials
ASTM D 256	-	Standard Test Methods for Impact Resistance of Plastics and Electrical Insulating Materials
ASTM D 257	-	Standard Test Methods for D-C Resistance or Conductance of Insulating Materials
ASTM D 570	-	Standard Test Methods for Water Absorption of Plastics
ASTM D 635	-	Standard Test Method for Rate of Burning and/or Extent of Time of Burning and Self-Supporting Plastics in a Horizontal Position
ASTM D 648	-	Standard Test Methods for Deflection Temperature of Plastics Under Flexural Load
ASTM D 696	-	Standard Test Methods for Coefficient of Linear Thermal Expansion of Plastics Between -30 Degrees and 30 Degrees C
ASTM D 790	-	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D 792	-	Standard Test Methods for Specific Gravity (Relative Density) and Density of Plastics by Displacement
ASTM D 955	-	Standard Test Methods of Measuring Shrinkage from Mold Dimensions of Molded Plastics
ASTM D 2240	-	Standard Test Methods for Indentation Hardness of Rubber Property Hardness

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ASTM D 3123	-	Spiral Flow of Low-Pressure Thermosetting Molding Compounds
ASTM D 3951	-	Commercial Packaging, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (6.2), a first article sample of the molding compound shall meet the requirements of this specification (see 4.3.1).

3.2 Material. The molding compound shall be a thermosetting mineral-filled epoxy resin formulated in granular form for application by the transfer molding process. The color of the molding compound after cure shall be black.

3.3 Electrical properties. The cured molding compound shall conform to the electrical requirements specified in Table I (see 4.4.2).

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TABLE I. Electrical properties of cured molding compound.

Property	Value
Dielectric strength, volts per mil at +20° Celsius (C), minimum	320
Dielectric constant from 100 Hertz (Hz) to 10 gigahertz (ghz), maximum	6.5
Volume resistivity, ohm-centimeter at +25°C, minimum	1×10^{13}
Dissipation factor from 100 Hz to 100 megahertz per second (mHz), maximum	0.06
Dissipation factor from 100 Hz to 10 gHz, maximum	0.05

3.4 Mechanical properties. The cured molding compound shall conform to the physical and mechanical properties specified in Table II (see 4.4.3).

TABLE II. Physical and mechanical properties of cured molding compound.

Property	Value
Flexural Strength, lbf/in^2, minimum	13,000
Izod impact, foot pounds per inch of notch, minimum	0.30
Hardness, Shore D, minimum	90
Deflection temperature, degrees C, minimum	+127
Coefficient of linear thermal expansion, inch per inch per 0.56°C, maximum:	
From +24 to +93°C	1.8×10^{-5}
From +93 to -149°C	3.3×10^{-5}

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TABLE II. Physical and mechanical properties of cured molding compound - Continued.

Property	Value
Coefficient of thermal conductivity, British thermal units per hour per square foot per 0.56°C per inch, minimum	4.2
Linear shrinkage of cold mold to cold part, inch per inch, maximum	0.008
Cure temperature, °C, minimum to maximum	+138, 20 minute maximum to +117, 20 minute maximum (+280 to +350)
Post-Cure, °C	149 ± 10
Moisture absorption, percent, maximum	0.5
Flammability	Self extinguishing
Specific gravity at +23°C, maximum	2.3
Spiral flow at +23°C and 1000 lbf/in^2, inches, minimum	30

3.5 Shelf life. The shelf life of the molding compound in unopened containers shall be not less than 6 months from date of manufacture when stored at temperatures from +10 to +20°C (see 4.4.4).

3.6 Workmanship. The workmanship shall be such as to insure a high quality product which is uniform, free of contaminants, and in conformance with this specification(see 3.2 and 4.4.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).

b. Quality conformance inspection (see 4.3.2).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in (applicable test method document or applicable paragraph(s) in the specification.

4.3.1 First article inspection. First article inspections shall consist of examinations and tests for the characteristics indicated in tables I, II and III.

4.3.2 Quality conformance inspection. Quality conformance inspections for acceptance of the molding compound shall consist of the following examinations and tests shown in table III.

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Detect Examine

TABLE III. Quality conformance inspection.

Characteristics	Requirement paragraph	Test paragraph
Hardness	table II	4.4.3.3
Specific gravity	table II	4.4.3.11
Spiral flow	table II	4.4.3.1.2
Workmanship	3.6	4.4.5

4.3.2.1 Lot size. Lot size shall consist of all the molding compound manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.3.2.2 Sampling. Unless otherwise specified (see 6.2), a sufficient quantity of molding compound from one unit representative of each lot (see 4.3.2.1) shall be selected at random for quality conformance testing. Each container of material shall be considered as a unit of product. Failure of the sample to meet quality conformance requirements shall be cause for lot rejection.

4.4 Test methods and procedures.

4.4.1 Test specimen preparation. The test specimen shall be prepared from molding compound in accordance with the manufacturer's recommendations.

4.4.2 Electrical properties.

4.4.2.1 Dielectric strength. The dielectric strength shall be determined in accordance with ASTM D 149. The dielectric strength shall be as specified in table I.

4.4.2.2 Dielectric constant. The dielectric constant shall be determined in accordance with ASTM D 150. The dielectric constant shall be as specified in table I.

4.4.2.3 Volume resistivity. The volume resistivity shall be determined in accordance with ASTM D 257. The volume resistivity shall be as specified in table I.

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4.4.2.4 Dissipation factor. The dissipation factor shall be determined in accordance with ASTM D 150. The dissipation factor shall be as specified in table I.

4.4.3 Mechanical properties.

4.4.3.1 Flexural strength. The flexural strength shall be determined in accordance with ASTM D 790. The flexural strength shall be as specified in table II.

4.4.3.2 Izod impact. The Izod impact resistance shall be determined in accordance with ASTM D 256, Method A. The Izod impact resistance shall be as specified in table II.

4.4.3.3 Hardness. The hardness shall be determined in accordance with ASTM D 2240. The hardness shall be as specified in table II.

4.4.3.4 Deflection temperature. The deflection temperature shall be determined in accordance with ASTM D 648. The deflection temperature shall be as specified in Table II.

4.4.3.5 Coefficient of linear thermal expansion. The coefficient of linear thermal expansion shall be determined in accordance with ASTM D 696. The coefficient of linear thermal expansion shall be as specified in table II.

4.4.3.6 Coefficient of thermal conductivity. The coefficient of thermal conductivity shall be determined in accordance with MIL-I-16923. The thermal conductivity shall be as specified in table II.

4.4.3.7 Linear shrinkage. The linear shrinkage of cold mold to cold part shall be determined in accordance with ASTM D 955. The linear shrinkage shall be as specified in table II.

4.4.3.8 Cure temperature. The cure temperature shall be measured at the geometric center of the test specimen during the entire curing cycle. The cure temperature shall be as specified in table II.

4.4.3.9 Moisture absorption. The percentage of moisture shall be determined in accordance with ASTM D 570, Procedure 6(a). The percentage weight gain shall be as specified in table II.

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4.4.3.10 Flammability. The flammability shall be determined in accordance with ASTM D 635. The molding compound shall be self-extinguishing (see table II).

4.4.3.11 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 792, Method A-1. The specific gravity shall be as specified in table II.

4.4.3.12 Spiral flow. The spiral flow shall be determined in accordance with ASTM D 3123. The spiral flow shall be as specified in table II.

4.4.4 Shelf life. The molding compound shall be tested for conformance with all requirements specified herein after completion of the storage period (see 3.5). The manufacturer's certificate of compliance with the shelf life requirements may be accepted during the storage period.

4.4.5 Workmanship. The molding compound shall be visually examined to determine compliance with the workmanship requirements of 3.6 (see 3.2).

4.5 Packaging inspection. The packaging and marking shall be inspected to determine compliance with Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C (see 6.2).

5.1.1 Level A. The molding compound shall be packaged in accordance with MIL-P-116, Method III, using containers conforming to PPP-C-96, PPP-D-729 or PPP-D-705.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. Preservation and packaging shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or C (see 6.2).

5.2.1 Level A. Containers conforming to PPP-C-96 shall be packed in snug-fitting fiberboard boxes which conform to PPP-B-636, type CF or SF, weather resistant; PPP-B-640, Class 2; PPP-B-601, style A or B; or PPP-B-621, Class 2, as applicable. Containers

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conforming to PPP-D-729 or PPP-D-705 shall not require over packing for shipment.

5.2.2 Level B. Same as Level A, except domestic style containers shall be used.

5.2.3 Level C. Packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Level A. Marking shall be in accordance with MIL-STD-129.

5.3.2 Level C. Marking shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The molding compound covered by this specification is intended for use in transfer molding applications requiring high overall strength as well as good thermal shock properties. In the use of the molding compound, consideration shall be given to the physical effects on the electronic devices.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. When a first article is required (see 3.1 and 4.3.1)

d. Sampling plan if other than specified (see 4.3.2.2)

e. Requirement for certification of shelf life (see 3.5 and 4.4.4)

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f. Applicable level of preservation, packaging, and packing (see 5.1 and 5.2)

g. Requirements for preservation, packaging, packing, and marking, if different (see 5.1, 5.2 and 5.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Encapsulating material, electrical
Electrical thermal conductivity
Electronic composite material
Embedding material, electrical
Potting material, electrical
Putty, electrical

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review:
DLA - GS

Preparing activity:
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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-47224B	930104

3. DOCUMENT TITLE
COMPOUND, MOLDING, TRANSFER, EPOXY RESIN, SINGLE-COMPONENT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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